Theorem, proposition, corollary, assumption, and section numbers refer to the main paper, "The Explosion Afraid of Itself."

A Full Proofs

A.1 Proof of Theorem 1

Under Assumption 1, the agent evaluates π_m by minimizing $\mathbf{G}$ over the augmented state space. The ambiguity component at a post-modification timestep τ is

$$\text{Ambiguity}(\pi_m, \tau) = \mathbb{E}_{Q(s_\tau^+ \mid \pi_m)}\big[H[\hat{P}(o_\tau \mid s_\tau^{env})]\big], \tag{1}$$

where $\hat{P}(o|s) = \int q(\tilde{m}) P_{\tilde{m}}(o|s)\, d\tilde{m}$ is the mixture under the agent's posterior $q(\tilde{m} \mid m, d_m)$ over successor parameters.

Step 1: Entropy decomposition. By the chain rule for entropy applied to the joint $(\tilde{m}, o)$ under q:

$$H[\hat{P}(o \mid s)] = \mathbb{E}_q\big[H[P_{\tilde{m}}(o \mid s)]\big] + I_q(o; \tilde{m} \mid s). \tag{2}$$

Under π_0, q is a point mass at m: $\mathbb{E}_q[H[P_{\tilde{m}}]] = H[P_m]$ and $I_q \equiv 0$. The excess ambiguity relative to π_0 therefore splits:

$$\text{Ambiguity}(\pi_m, \tau) - \text{Ambiguity}(\pi_0, \tau) = \underbrace{\big(\mathbb{E}_q H[P_{\tilde{m}}] - H[P_m]\big)}_{\text{intrinsic}} + \underbrace{I_q(o; \tilde{m} \mid s)}_{\text{modification-induced}}. \tag{3}$$

Step 2: Geometric floor on the MI term. The Fisher information matrix is the Hessian of KL at the basepoint, giving $D_{KL}[P_m \| P_{\tilde{m}}] = \frac{1}{2}\, d_F^2(\mathbf{A}, \tilde{\mathbf{A}}) + O(d_F^3)$ locally. The agent's posterior q at the rational-allocation optimum has Fisher-Rao spread bounded below by the modification magnitude—the residual the agent leaves uncorrected when allocating finite simulation effort, forced by the σ^* optimization of Sect. 3.4 to scale with d_m rather than collapse to a point mass. Under Assumption 4, the local mutual information then satisfies

$$\mathbb{E}_s[I_q(o; \tilde{m} \mid s)] \;\geq\; C(\bar{\lambda}, \varepsilon) \cdot d_F^2(\mathbf{A}, \tilde{\mathbf{A}}) + O(d_F^3), \tag{4}$$

a local lower bound, with $C(\bar{\lambda}, \varepsilon)$ determined by the curvature analysis of Proposition 1. The likelihood $P_{\tilde{m}}(o|s)$ depends on $\mathbf{A}$ alone, so only the perceptual component of the modification enters the mutual information. This establishes Part (i).

Step 3: From ambiguity to EFE. Under the conditions of Part (ii), the intrinsic term satisfies $\mathbb{E}_q H[P_{\tilde{m}}] - H[P_m] \geq 0$, and the risk excess satisfies $\text{risk}(\pi_m, \tau) - \text{risk}(\pi_0, \tau) \geq 0$. Combining:

$$\mathbf{G}(\pi_m, \tau) - \mathbf{G}(\pi_0, \tau) \geq I_q(o; \tilde{m} \mid s) \geq g(d_m). \tag{5}$$

Summing over the post-modification horizon yields $\mathbf{G}(\pi_m) - \mathbf{G}(\pi_0) \geq (T - t_{mod}) \cdot g(d_m)$, establishing Part (ii). $\square$

A.2 Proof of Proposition 2 (Identifiability Barrier)

As $\lambda_{\min}(\mathbf{F}(\tilde{m})) \to 0$, at least one direction in parameter space becomes unidentifiable: the Cramér-Rao inequality forces the variance of any estimator of that parameter to diverge as $1/\lambda_{\min}$, so the agent's posterior $q(\tilde{m})$ acquires unbounded spread along the degenerate direction. When that direction lies in the likelihood — the perceptual parameters $\mathbf{A}$, which the ambiguity channel tracks — the mutual information $I_q(o; \tilde{m} \mid s)$ diverges; this is the case relevant to escaping the unconditional brake. By Theorem 1(i), the per-timestep ambiguity cost diverges, and the total expected free energy over any finite horizon diverges with it. The transition from an identifiable to an unidentifiable representation is therefore maximally costly under the agent's own objective. □

A.3 Proof of Theorem 2

At each timestep t of the recursive trajectory, the evaluating agent m_0 predicts post-modification observations under m_t. The prediction is a tangent-space extrapolation from m_0, not from m_{t-1}: the evaluation is performed by m_0 using m_0's representational capacity, regardless of how many modifications intervene. Theorem 1(i) applied with m_0 as base and m_t as target therefore gives, at each timestep $t \geq 1$:

$$\mathbb{E}_s\big[I_{q_t}(o; m_t \mid s)\big] \geq g(d_F(\mathbf{A}_0, \mathbf{A}_t)). \tag{6}$$

Summing over t:

$$\sum_{t=1}^{T} \mathbb{E}_s\big[I_{q_t}(o; m_t \mid s)\big] \geq \sum_{t=1}^{T} g(d_F(\mathbf{A}_0, \mathbf{A}_t)). \tag{7}$$

For trajectories satisfying $d_F(m_0, m_{t+1}) \geq d_F(m_0, m_t)$ — the threat case of unidirectional modification — the per-step penalty is non-decreasing, and $f_{t+1} = g(d_F(m_0, m_{t+1})) \geq g(d_F(m_0, m_t)) = f_t$. For trajectories in which the agent retreats, the bound weakens proportionally to the reduction in endpoint distance, which is the correct behavior: a trajectory returning to m_0 has performed no net self-modification and should not incur perpetual cost. □

A.4 Proof of Corollary 1 (Alignment Drift Bound)

Part (i): The model-to-policy map $m \mapsto \pi(m)$ passes through two stages: the expected free energy map $m \mapsto G(m)$ and the softmax policy selection $G \mapsto \pi = \sigma(-\gamma G)$.

The softmax Jacobian $\partial\pi/\partial G$ has spectral norm bounded by $\gamma/4$ (the derivative of the logistic function is maximized at $1/4$, scaled by precision γ).

The expected free energy gradient, measured in the Fisher dual norm, satisfies $\|\nabla_\theta G\|_{F^{-1}} \leq K$ for a constant K depending on model structure. The composite map from geodesic distance to TV distance between policies therefore satisfies:

$$\delta_{align} \leq L_{FR} \cdot d_m, \quad L_{FR} \leq \gamma/2 \tag{8}$$

Inverting: any target δ_{align} requires $d_m \geq \delta_{align}/L_{FR}$. By Theorem 1, the expected free energy cost is at least $g(\delta_{align}/L_{FR}) \geq C \cdot (\delta_{align}/L_{FR})^2$. With $L_{FR} \leq \gamma/2$:

$$\Delta\mathbf{G} \geq \frac{4C}{\gamma^2} \cdot \delta_{align}^2 \tag{9}$$

The cost is quadratic in alignment degradation, with computable coefficient.

For modifications leaving $\mathbf{A}$ fixed the composite map factors through the risk channel of Section 3.4; the same quadratic bound holds with C replaced by C_{risk} and L_{FR} replaced by ℓ_{FR}^{-1}.

Part (ii): Near a bifurcation, the Jacobian of the policy map has at least one eigenvalue approaching infinity. The agent evaluating a modification near such a point must propagate its parameter uncertainty through this near-singular map. Small uncertainty in d_m produces large uncertainty in the successor's policy — high entropy in the predictive distribution over post-modification behavior. The ambiguity cost diverges as the bifurcation point is approached. □

B Computational Validation: Details

This appendix documents the methods, results, and design choices underlying Sect. 4.4. The code reproduces every figure and table and is available as supplementary material. All draws are seeded; identical inputs produce identical outputs.

B.1 POMDP Construction

Random POMDPs are drawn with $n_o = n_a = n_s$. The likelihood-only measurements (perceptual floor and its compounding) use $n_s \in \{3, 5, 8, 12\}$. The expected-free-energy measurements (the two-channel grid, the risk channel, the desperate-agent boundary) use $n_s \in \{3, 5, 8\}$: policy enumeration is $n_a^\tau = n_s^2$ at $\tau = 2$, so foreseen-risk evaluation, which dominates runtime, grows with n_s; the likelihood-only sweeps carry no policy enumeration and run cheaply to $n_s = 12$. Free parameters (each k-simplex contributes $k - 1$ degrees of freedom) range from 28 at $n_s = 3$ to 1738 at $n_s = 12$.

Each simplex-valued parameter (columns of $\mathbf{A}$, columns of $\mathbf{B}$ per action, and the vectors $\mathbf{C}$, $\mathbf{D}$) is drawn from a symmetric Dirichlet distribution with concentration $\alpha = 0.5$, then floored at $p_i \geq 0.05$ and renormalized to keep trajectories interior. Sharp models ($\alpha = 0.5$) are used throughout: at $\alpha = 1.5$ the soft expected-free-energy policy is nearly uniform and the risk channel fails to engage (TV distance between $\pi^*(m_0)$ and $\pi^*(\tilde{m})$ for a $d_F = 0.5$ perturbation falls to 0.04). Every expected-free-energy computation uses policy precision $\gamma = 16$, which produces discriminating policies (EFE range 2.36–2.69) while holding the construction-noise floor low; higher γ sharpens π^* but amplifies that floor. The perceptual-floor and compounding measurements act on the likelihood alone and are independent of γ.

The non-desperate preference vector $\mathbf{C}$ is set one-shot from the achieved observation distribution of the agent's policy at m_0, which leaves a residual foreseen_risk$(m_0, m_0) \sim 10^{-3}$ rather than zero. Standard active inference treats $\mathbf{C}$ as exogenous, so $\mathbf{C}$ is not iterated to a self-consistent fixed point; the $\sim 10^{-3}$ residual is the construction's noise floor and is reported explicitly (Sect. B.6).

B.2 Fisher-Rao Geometry and Trajectory Construction

The Fisher-Rao distance on the product manifold decomposes as

$$d_F(m, \tilde{m}) = \sqrt{\sum_i d_F^2(p_i, \tilde{p}_i)}, \tag{10}$$

summing over every simplex-valued factor p_i. For each categorical factor the geodesic distance is $d_F(p, q) = 2 \arccos(\sum_j \sqrt{p_j q_j})$, the arc length on a sphere of radius 1/2 [1,2]. It is computed in the equivalent Hellinger form $d_F = 4 \arcsin(H/\sqrt{2})$ with $H^2 = \frac{1}{2} \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$, which differences $\sqrt{p} - \sqrt{q}$ directly and so returns exactly zero for identical inputs; the arccos form loses precision near identity (a simplex summing to one ULP below 1 yields $\arccos(0.999\ldots) \approx 2 \times 10^{-8}$ rather than 0). Geodesics use the $\sqrt{p}$-parameterization on the unit sphere with spherical interpolation (slerp). Because the categorical manifold is isometric to the positive orthant of that sphere, a slerp step along an unlucky tangent can leave the orthant and undershoot the requested arc; tangent directions are rejection-sampled (up to 20 retries) to keep the step interior, with up to 20% shortfall tolerated for columns near the boundary.

Four trajectory geometries are tested:

(a) Fisher-Rao geodesic. Per-step slerp in $\sqrt{p}$-coordinates along a random unit tangent at m_0; $d_F(m_0, m_t) = t\delta$ holds by construction to float precision.

(b) Parameter-space line. A control: the straight segment $m_0 \to m_T$ in simplex coordinates. Slopes agree with the geodesic to two decimal places, and values at matched d_F differ by under 12% (a higher-order curvature correction).

(c) Uncorrelated random walk. A fresh random tangent is sampled at the current model each step and a slerp step of arc length δ applied.

(d) Correlated random walk. The fresh tangent is mixed with a parallel-transported drift direction, $u_t = \rho \cdot \mathrm{PT}(u_{\mathrm{drift}}; m_0 \to m_t) + \sqrt{1-\rho^2} \cdot u_{\mathrm{fresh}}$, with $\rho = 0.5$ and parallel transport under the Levi-Civita connection on the unit sphere (per factor).

Endpoint-distance growth fits $d_F(m_0, m_t) \propto t^\alpha$ with $\alpha \approx 0.50$ (uncorrelated) and $\alpha \approx 0.73$ (correlated) across all sizes. The capability-driven (empowerment-ascending) trajectory is treated separately in Sect. B.7 as a boundary experiment, distinct from these four.

Table 1. Per-step ambiguity slope vs d_F^A across four trajectory geometries. The zeroth-order proxy $\mathbb{E}_s[\mathrm{KL}(\hat{A}\|A)]$ collapses to the theorem's quadratic floor (slope $\approx$ 2) across all trajectory types; the sample-based mutual-information estimator carries the next-order term (slope $\approx$ 4). Fits at $n_s \in \{3, 5, 8, 12\}$, 25 seeds per cell.

Trajectory	slope (KL)	R^2	slope (MI)	R^2
Geodesic	2.08	0.84	4.43	0.92
Parameter line	2.06	0.86	4.41	0.92
Pure RW	2.05	0.72	4.88	0.88
Correlated RW (ρ=0.5)	2.01	0.76	4.56	0.88

B.3 Estimator Choice, the Perceptual Floor, and Path-Independence

The bound of Proposition 1 is a floor on residual mutual information at the basepoint. The matched empirical quantity is the KL divergence between the agent's basepoint estimate and the actual successor likelihood, computed with the zeroth-order estimator $\hat{m} = m_0$ — the tangent-space estimator at zero compute spend, which is the estimator the Toponogov bound applies to directly. Across all four geometries this zeroth-order proxy collapses onto a slope-2 envelope against d_F^A (Table 1). The sample-based mutual-information estimator carries the next-order term at slope $\approx$ 4.4–4.9, sitting above the floor: the realized cost grows faster than the bound, never below it.

Higher-order estimators (a first-order Taylor step in parameter space, or in $\sqrt{p}$-coordinates) reduce the residual but incur compensating compute cost. Under the agent's free-energy calculus (Sect. 3.4) the total cost (residual plus simulation effort) cannot fall below the zeroth-order floor, so the zeroth-order proxy is reported throughout the main body because it is the quantity the bound governs.

The collapse is path-independent. The log-log slope is ≈ 2 for every geometry; the spread across geometries (2.01–2.08) and the mild variation in fitted coefficient reflect higher-order curvature on the product manifold — different paths reach m_t with slightly different per-component d_F breakdowns, altering the coefficient of $(d_F^A)^2$ without changing the exponent — rather than a scaling violation.

B.4 Cumulative Compounding

Cumulative zeroth-order ambiguity over a T-step trajectory follows the geometry-set spread of Theorem 2 (Table 2). Each fit sits below its asymptote by the finite-T truncation of $\sum_{t=1}^{T} t^{2\alpha}$ at $T = 10$: the geodesic, with $d_F(m_0, m_t) = t\delta$, approaches T^3; the correlated walk approaches $T^{2.5}$; the uncorrelated walk approaches T^2.

Table 2. Cumulative compounding slopes for the zeroth-order ambiguity proxy $\mathbb{E}_s[\mathrm{KL}(\hat{A}_t \| A_t)]$ against trajectory length T. Fits at $n_s \in \{3, 5, 8, 12\}$, 25 seeds per cell, $T_{\max} = 10$.

Trajectory	fit slope	R^2	predicted asymptote
Geodesic	2.62	1.00	T^3
Correlated RW (ρ=0.5)	2.02	1.00	$T^{2.5}$
Pure RW	1.75	1.00	T^2

B.5 Two-Channel Structure and Calibration

The single-step grid crosses magnitude (trivial 0.05, radical 0.6) $\times$ component ($\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$, blended) $\times$ regime (accurate, inaccurate, desperate), with 20 seeds at $n_s \in \{3, 5, 8\}$ and $\gamma = 16$. Per cell it records foreseen ambiguity, foreseen risk excess, realized risk excess, and the foreseen$-$realized gap.

Ambiguity is perception-only. $\mathbf{B}/\mathbf{C}/\mathbf{D}$-only perturbations register foreseen ambiguity at machine epsilon ($\sim 10^{-16}$); only $\mathbf{A}$-only and blended perturbations carry positive ambiguity, scaling with d_F^A.

Risk is magnitude-monotone, dominated by preference. Foreseen risk excess is positive for every component and grows from trivial to radical (Table 3), dominated by preference ($\mathbf{C}$) by an order of magnitude; dynamics ($\mathbf{B}$) and priors ($\mathbf{D}$) are weak. The per-component log-log slope of foreseen risk against d_F sits between 1 and 2 ($\approx$1.0–1.2): risk is magnitude-monotone, but in the finite-policy soft-EFE setup it is not strictly quadratic in d_m. The quadratic is the theorem's lower bound, not the measured exponent. The per-component risk also compounds: a T-step per-component chain (12 seeds) yields a cumulative foreseen-risk slope $\approx$ 1.7–1.8, consistent with magnitude-monotone but sub-quadratic per-step risk.

Calibration gap. For accurate priors, foreseen equals realized by construction and the gap is ≈ 0. For inaccurate priors the gap is sign-mixed and grows with magnitude: corrective modifications, those toward the world, read as foreseen-costly while lowering realized risk. This is the empirical content of the brake guarding the agent's model rather than the truth (Sect. 6.3, and the calibration discussion of the main text).

B.6 The Desperate-Agent Boundary

The desperate regime is *foreseen*-desperate: $\mathbf{C}$ is drawn independently of the achievable observation distribution, so the agent's own model foresees failing to reach its preferences and the Theorem 1(ii) condition breaks. This tests the Sect. 5 boundary directly. Baseline foreseen risk foreseen_risk(m_0, m_0) is $\sim 10^{-3}$ for non-desperate regimes and ~ 3.7–4.2×10^{-1} for the desperate regime (Table 4). The foreseen-risk reduction a random radical modification can secure —

Table 3. Mean foreseen risk excess (in nats) across the single-step grid: regime × magnitude × component. Accurate and inaccurate priors are constructed foreseen-non-desperate; the desperate regime breaks the Theorem 1(ii) condition. Cells averaged across 20 seeds and three model sizes ($n_s \in \{3, 5, 8\}$).

Regime	Mag.	A	B	C	D	blended
accurate	trivial	-5.34×10^{-5}	-4.54×10^{-6}	1.90×10^{-4}	1.81×10^{-6}	2.50×10^{-5}
accurate	radical	-3.93×10^{-4}	1.70×10^{-5}	6.78×10^{-3}	3.40×10^{-4}	8.54×10^{-4}
inaccurate	trivial	-1.04×10^{-4}	1.73×10^{-5}	4.51×10^{-5}	2.71×10^{-6}	3.00×10^{-5}
inaccurate	radical	-1.82×10^{-4}	2.90×10^{-4}	5.96×10^{-3}	-1.05×10^{-4}	1.55×10^{-3}
desperate	trivial	-1.86×10^{-4}	-1.24×10^{-6}	4.37×10^{-5}	9.10×10^{-5}	-1.53×10^{-4}
desperate	radical	2.02×10^{-3}	8.58×10^{-4}	1.25×10^{-2}	4.28×10^{-3}	6.92×10^{-4}

Table 4. Brake floor vs. baseline foreseen risk, by regime and model size. Baseline = foreseen_risk(m_0, m_0); floor = $\min_{\tilde{m}}$ foreseen_risk($m_0, \tilde{m}$) over random blended perturbations at radical magnitude. The available reduction is baseline−floor; for non-desperate regimes this equals the construction's soft-policy noise floor ($\sim 10^{-3}$), for the desperate regime it is the foreseen brake relaxation ($\sim 10^{-1}$).

Regime	n_s	baseline	floor	reduction
accurate	3	1.13×10^{-2}	9.00×10^{-6}	1.13×10^{-2}
accurate	5	4.58×10^{-3}	1.73×10^{-4}	4.41×10^{-3}
accurate	8	1.65×10^{-3}	1.29×10^{-5}	1.63×10^{-3}
inaccurate	3	8.36×10^{-3}	4.42×10^{-5}	8.31×10^{-3}
inaccurate	5	4.99×10^{-3}	4.40×10^{-4}	4.55×10^{-3}
inaccurate	8	1.96×10^{-3}	2.44×10^{-5}	1.94×10^{-3}
desperate	3	3.91×10^{-1}	3.07×10^{-2}	3.61×10^{-1}
desperate	5	4.23×10^{-1}	4.63×10^{-2}	3.77×10^{-1}
desperate	8	3.70×10^{-1}	1.60×10^{-1}	2.10×10^{-1}

baseline minus the floor — is $\sim 10^{-3}$ for non-desperate agents, equal to the construction's noise floor and therefore no real signal, against $\sim 10^{-1}$ for the desperate agent. The brake relaxes by 30–130× in absolute scale, keyed to the agent's own foreseen failure rather than to any external signal. It does not vanish: across sizes the desperate floor stays in the 10^{-2}–10^{-1} band (3.07×10^{-2} at $n_s = 3$ up to 1.60×10^{-1} at $n_s = 8$, Table 4), so the brake slackens without releasing. Relative reduction is $\sim$95% in both regimes and is therefore misleading; the absolute scale is the discriminating quantity.

B.7 Empowerment Boundary

An additional experiment lies outside the four-geometry headline - an "empowerment maximizing agent". A capability-driven trajectory ascends $\tau = 2$ empowerment by best-of-N selection: at each step $N = 80$ candidate tangents are sampled, a slerp step of arc $\delta = 0.05$ applied to each, and the candidate maximizing empowerment selected. The empowerment metric is $\mathbb{E}_{s_0 \sim D}[I(a_{1:2}; o_2 \mid s_0)]$

under a uniform prior on action sequences, a computable lower bound on channel capacity. Run for 15 seeds at $n_s \in \{3, 5, 8\}$, $T = 10$.

The ascent concentrates modification in $\mathbf{B}$ (transitions, which carry no ambiguity cost): at $n_s = 8$ the $\mathbf{B}$-fraction of d_F is 0.87 against an $\mathbf{A}$-fraction of 0.46. Foreseen ambiguity stays at 10^{-10}–10^{-7} — empowerment walks around the ambiguity brake — while foreseen risk grows from 10^{-6} to 10^{-4}: empowerment still pays the risk channel. This demonstrates the boundary of the theorems. An active-inference agent is braked from these $\mathbf{B}$-concentrated modifications by the risk channel; a pure empowerment maximizer overrides the ambiguity brake.

B.8 Reproduction

Complete code — POMDP construction, trajectory generators, estimators, expected-free-energy evaluation, and plotting — is provided as supplementary material, pinned to NumPy 2.4.6, SciPy 1.17.1, and Matplotlib 3.10.9 under Python 3.12. The full suite (headline $n_s \in \{3, 5, 8, 12\}$, 25 seeds; grid $n_s \in \{3, 5, 8\}$, 20 seeds; recursive, 12 seeds; empowerment, 15 seeds; $T = 10$) runs in $\sim$135 s on a single CPU thread. All RNG draws are seeded, so identical inputs reproduce identical outputs. A suite of 49 unit tests covers the geometric primitives, perturbation selectivity, the expected-free-energy properties, the foreseen/realized identities, the slope-2 floor, the desperate-agent construction, and determinism.

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